```
// ====================================
// COLLEGE CLUB MANAGEMENT SYSTEM
// ====================================

// Default Club Data

const defaultClubs = [
  {
    id: 1,
    name: "Coding Club",
    category: "Technology",
    description: "Learn programming, web development and software skills.",
    members: []
  },
  {
    id: 2,
    name: "Photography Club",
    category: "Creative",
    description: "Explore photography, editing and creative activities.",
    members: []
  },
  {
    id: 3,
    name: "Sports Club",
    category: "Sports",
    description: "Participate in sports competitions and fitness events.",
    members: []
  },
  {
    id: 4,
    name: "Music Club",
    category: "Arts",
    description: "Discover music, singing and musical instruments.",
    members: []
  },
  {
```

```
    id: 5,
    name: "Science Club",
    category: "Education",
    description: "Explore science projects and technical innovations.",
    members: []
  },
  {
    id: 6,
    name: "Cultural Club",
    category: "Culture",
    description: "Celebrate cultural activities and college festivals.",
    members: []
  }
];

const defaultEvents = [
  {
    id: 1,
    name: "Web Development Workshop",
    date: "2026-12-10",
    venue: "Computer Lab",
    registrations: []
  },
  {
    id: 2,
    name: "Photography Competition",
    date: "2026-12-15",
    venue: "College Auditorium",
    registrations: []
  },
  {
    id: 3,
    name: "Sports Day",
    date: "2026-12-20",
    venue: "College Ground",
    registrations: []
```

```
  }
];

// Load and Save Data

function loadData(key, defaultValue) {
  try {
    const saved = localStorage.getItem(key);
    return saved ? JSON.parse(saved) : defaultValue;
  } catch (error) {
    console.error("Data loading error:", error);
    return defaultValue;
  }
}

let clubs = loadData("collegeClubs", defaultClubs);
let students = loadData("collegeStudents", []);
let events = loadData("collegeEvents", defaultEvents);

function saveData() {
  localStorage.setItem("collegeClubs", JSON.stringify(clubs));
  localStorage.setItem("collegeStudents", JSON.stringify(students));
  localStorage.setItem("collegeEvents", JSON.stringify(events));
}

// Prevent HTML injection when displaying user input

function escapeHTML(value) {
  return String(value).replace(/[&<>"']/g, character => ({
    "&": "&amp;",
    "<": "&lt;",
    ">": "&gt;",
    '"': "&quot;",
    "'": "&#039;"
  })[character]);
}
```

```
// Dashboard

function updateDashboard() {

  document.getElementById("clubCount").textContent =
    clubs.length;

  document.getElementById("memberCount").textContent =
    clubs.reduce((total, club) ⇒
      total + club.members.length, 0);

  document.getElementById("eventCount").textContent =
    events.length;

  document.getElementById("registrationCount").textContent =
    events.reduce((total, event) ⇒
      total + event.registrations.length, 0);
}

// Display Clubs

function displayClubs(searchText = "") {

  const clubList = document.getElementById("clubList");

  const filteredClubs = clubs.filter(club ⇒
    club.name.toLowerCase().includes(searchText.toLowerCase()) ||
    club.category.toLowerCase().includes(searchText.toLowerCase())
  );

  if (filteredClubs.length === 0) {
    clubList.innerHTML = "<p>No clubs found.</p>";
    return;
  }
```

```
  clubList.innerHTML = filteredClubs.map(club => `
    <div class="club-card">

      <h3>🎓 ${escapeHTML(club.name)}</h3>

      <p><strong>Category:</strong>
      ${escapeHTML(club.category)}</p>

      <p>${escapeHTML(club.description)}</p>

      <p><strong>Members:</strong>
      ${club.members.length}</p>

      <button
        onclick="joinClub(${club.id})"
        ${club.members.length === students.length && students.length > 0
          ? "disabled" : ""}>
        Join Club
      </button>

    </div>
  `).join("");
}

// Search Clubs

document.getElementById("clubSearch")
.addEventListener("input", function () {
  displayClubs(this.value);
});

// Student Registration

document.getElementById("studentForm")
.addEventListener("submit", function(event) {
```

```
event.preventDefault();

const name = document.getElementById("studentName").value.trim();
const roll = document.getElementById("rollNumber").value.trim();
const email = document.getElementById("studentEmail").value.trim();
const department = document.getElementById("department").value.trim();

if (!name || !roll || !email || !department) {
  alert("Please fill all fields.");
  return;
}

const duplicate = students.some(student =>
  student.roll.toLowerCase() === roll.toLowerCase() ||
  student.email.toLowerCase() === email.toLowerCase()
);

if (duplicate) {
  alert("Student already registered!");
  return;
}

const student = {
  id: Date.now(),
  name,
  roll,
  email,
  department
};

students.push(student);

saveData();

displayStudents();
updateDashboard();
```

```
    displayClubs();

    this.reset();

    alert("Student registered successfully!");
});

// Display Students

function displayStudents() {

    const studentList = document.getElementById("studentList");

    studentList.innerHTML = students.map(student => `
        <tr>
            <td>${escapeHTML(student.name)}</td>
            <td>${escapeHTML(student.roll)}</td>
            <td>${escapeHTML(student.email)}</td>
            <td>${escapeHTML(student.department)}</td>
        </tr>
    `).join("");

    if (students.length === 0) {
        studentList.innerHTML =
            "<tr><td colspan='4'>No students registered.</td></tr>";
    }
}

// Join Club

function joinClub(clubId) {

    if (students.length === 0) {
        alert("Please register as a student first.");
        return;
    }
```

```
  const roll = prompt("Enter your Register Number:");

  if (!roll) return;

  const student = students.find(s ⇒
    s.roll.toLowerCase() === roll.trim().toLowerCase()
  );

  if (!student) {
    alert("Student not found. Please register first.");
    return;
  }

  const club = clubs.find(c ⇒ c.id === clubId);

  if (!club) return;

  if (club.members.includes(student.id)) {
    alert("You have already joined this club!");
    return;
  }

  club.members.push(student.id);

  saveData();

  displayClubs();
  updateDashboard();

  alert("Successfully joined " + club.name + "!");
}

// Create New Event

document.getElementById("eventForm")
```

```
.addEventListener("submit", function(event) {

    event.preventDefault();

    const name = document.getElementById("eventName").value.trim();
    const date = document.getElementById("eventDate").value;
    const venue = document.getElementById("eventVenue").value.trim();

    if (!name || !date || !venue) {
        alert("Please fill all event details.");
        return;
    }

    if (date < new Date().toLocaleDateString("en-CA")) {
        alert("Please select a future date.");
        return;
    }

    const newEvent = {
        id: Date.now(),
        name,
        date,
        venue,
        registrations: []
    };

    events.push(newEvent);

    saveData();

    displayEvents();
    updateDashboard();

    this.reset();

    alert("New event created successfully!");
```

```
});

// Display Events

function displayEvents() {

  const eventList = document.getElementById("eventList");

  if (events.length === 0) {
    eventList.innerHTML = "<p>No events available.</p>";
    return;
  }

  eventList.innerHTML = events.map(item => `
    <div class="event-card">

      <h3>📅 ${escapeHTML(item.name)}</h3>

      <p><strong>Date:</strong>
      ${escapeHTML(item.date)}</p>

      <p><strong>Venue:</strong>
      ${escapeHTML(item.venue)}</p>

      <p><strong>Registrations:</strong>
      ${item.registrations.length}</p>

      <button onclick="registerEvent(${item.id})">
        Register for Event
      </button>

    </div>
  `).join("");
}

// Register for Event
```

```
function registerEvent(eventId) {

  if (students.length === 0) {
    alert("Please register as a student first.");
    return;
  }

  const roll = prompt("Enter your Register Number:");

  if (!roll) return;

  const student = students.find(s ⇒
    s.roll.toLowerCase() === roll.trim().toLowerCase()
  );

  if (!student) {
    alert("Student not found!");
    return;
  }

  const selectedEvent = events.find(e ⇒ e.id === eventId);

  if (!selectedEvent) return;

  if (selectedEvent.registrations.includes(student.id)) {
    alert("You have already registered for this event!");
    return;
  }

  selectedEvent.registrations.push(student.id);

  saveData();

  displayEvents();
  updateDashboard();
```

```
    alert("Event registration successful!");
}

// Initial Application Loading

function initializeApp() {
    displayClubs();
    displayStudents();
    displayEvents();
    updateDashboard();
}

initializeApp();
```